



UNIVERSITY OF AMSTERDAM

CLINGENDAEL INSTITUTE

Gaining new insights into protests using AI

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Abstract

This project, conducted in collaboration with the Clingendael Institute, leverages artificial intelligence to gain deeper insights into the causes and dynamics of protests across Europe. Using a filtered version of the ACLED dataset, we developed models to categorize protest topics and predict the likelihood of violent escalation. The report explores two approaches to topic labeling: trigger-word-based and supervised machine learning. Supervised transformer models (BERT, RoBERTa, and T5) achieved high accuracy scores, with T5 showing the best generalization. For violence prediction, both binary classifiers and probability-based models were evaluated. Binary models had limited success due to class imbalance, but a probability model provided more nuanced risk assessments. While the system achieved high performance in topic classification, violence prediction remains challenging due to data skewness and given features. The report concludes with a discussion of ethical implications and recommendations for future work, including improving generalizability, expanding topic coverage, and incorporating political context.



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1 Introduction

1.1 The Clingendael Institute

The Clingendael Institute is a leading Dutch think tank focused on international relations, strategic policy, and geopolitical developments. Known for its independence and unbiased stance, Clingendael provides research and policy advice to governments, international institutions, and the public. One of its core missions is to strengthen European resilience by analyzing the impact of international developments on Dutch and EU security and stability (Clingendael 2025).

Clingendael is currently conducting research into protests and public dissent across Europe. In the context of rising political polarization and global tensions, the Palestine-Israel conflict, climate protests, and labor tensions, understanding the dynamics behind mass mobilization has become essential. They aim to identify both the topics that drive people to protest and the risk factors that lead to violent escalation. To support this effort, Clingendael has launched a challenge to leverage artificial intelligence (AI) to analyze protest data.

1.2 Problem Statement

This research focuses on two central questions:

How can protests across Europe be automatically classified according to their main topics, such as climate change, labor rights, or geopolitical conflicts? What factors contribute to the escalation of protests into violence, and how can models predict such violent outcomes despite their low occurrence?

1.3 Product Vision

The objective of this product is to provide Clingendael with data-driven insights into the nature, themes, and potential escalation of protests across Europe. Specifically, the tool is designed to automatically classify protest topics, identify temporal and geographic trends, and analyze structural factors that may influence the likelihood of a protest becoming violent.

According to Clingendael, the highest priority is the automated classification of protests by topic. This functionality supports their efforts to understand which societal and political issues are most notable across different regions and time periods. For instance, the system aims to capture how frequently protests address topics such as climate change, education, geopolitical conflicts (e.g., Palestine), or labor rights. By categorizing protests based on thematic content, Clingendael can better identify patterns of public discontent and political engagement.

A secondary, though still important, objective is to explore the relationship between various protest characteristics -such as location, date, actors involved, or protest size- and the likelihood of escalation into violence. This includes examining whether the presence of law enforcement correlates with protest outcomes, and whether certain protest topics are more prone to violent developments than others.

Ultimately, the product is intended to support efforts to maintain public order by helping stakeholders, including Dutch law enforcement, understand how protests evolve and what factors contribute to peaceful or violent outcomes. For example, by comparing similar protests across countries, law enforcement may identify effective strategies for de-escalation based on less violent outcomes observed elsewhere.

2 Data

2.1 ACLED dataset

The client's primary data source is the ACLED (Armed Conflict Location and Event Data) dataset, a global real-time database that tracks violent conflicts and protests (ACLED 2025). The Clingendael team has shared a filtered version of this dataset, spanning from January 1, 2018 to May 23, 2025, and focused exclusively on Europe. This subset includes 621,877 events with structured attributes -such as location, date, and type of event- and text in the form of "notes" describing the nature of each protest. In total there are 29 features (columns).

2.2 Preprocessing

Despite the dataset being edited by Clingendael, some irrelevant data still remains. Therefore, a bit of preprocessing is needed. Firstly, the dataset contains various types of events, such as explosions and battles. To make sure the data aligns with our focus, the "type of events" column has been filtered to include only protests, violence against civilians, and riots. Additionally, entries from Ukraine have been excluded due to the ongoing war, as their inclusion would heavily skew the dataset toward Ukrainian data. This removal has been requested on behalf of Clingendael. Duplicate rows have also been removed. Finally, rows without notes have been excluded, as the notes are essential for labeling.

3 Protest labeling

3.1 Implementation

3.1.1 Labeled data for training

To label protest topics, we extract them from text-based notes using a language model. While a hard-coded trigger-word approach lacks robustness, it can help create an initial labeled dataset and uncover new topic categories. Labeling starts by adding a "topic" column and using a script to surface untagged notes for manual review. Annotators search for topic-specific keywords (e.g., "fossil fuels" for "climate change"). Topics and their triggers are stored in a dictionary, which is updated as new terms are found and applied to the dataset in iterative passes.¹

Some ambiguity arises due to the overlap of trigger words across multiple topics. For example, "equal pay" may relate to both labor rights and women's rights. To address such cases, the order of topics in the dictionary is prioritized, ensuring that more specific or semantically dominant topics (e.g., women's rights) are matched before more general ones (e.g., labor rights). This can be done by either randomly or selectively reviewing notes with a corresponding topic. When a wrong label is found, it is up to the reviewer to find the underlying problem.

Importantly, the goal of this process is not complete coverage, but high-precision labeling to support supervised training. In collaboration with Clingendael, a minimum threshold of 500 labeled instances per topic was established to ensure adequate data for model training. Given the reliance on trigger words, some degree of labeling error is inevitable. While highly specific trigger words would reduce false positives, they are often too rare or time-consuming to identify. To assess the overall quality of the labeling, a random sample is selected and manually verified. The required sample size is determined using the standard sample size formula (Ahmed 2024)²:

$$n = \frac{z^2 \cdot p \cdot (1 - p)}{e^2}$$

In which Z represents the Z-score, p represents the expected proportion, and e represents the margin of error.

3.1.2 Machine learning

We fine-tuned different pre-trained transformer models to classify protest classes, in an attempt to preserve low computing power and high time efficiency. Transformer models are a unique kind of neural network that uses mathematical techniques, such as attention or self-attention, to learn the context and relationships in data (Vaswani et al. 2017). To maximize the potential for success in solving this labeling problem, the use of multiple transformer models is necessary. The three transformer models we used are *BERT*, *RoBERTa*, and *T5*. BERT was introduced as a deep learning model by Google Research in 2018 (Devlin et al. 2018). RoBERTa is a follow-up to BERT, developed by Meta AI. The main difference between RoBERTa and BERT is that RoBERTa is trained on a larger dataset (Yinhan Liu 2019). What differentiates T5 from the BERT models is its ability to handle multiple NLP tasks without needing to be fine-tuned. This

¹The final dictionary can be found in the appendix: Figure 6

²Since the outcomes are a very small amount in comparison to the total amount of rows, there is no need for a finite population correction.

model is trained on the C4 dataset, which consists of a large, clean corpus and is also developed by Google Research in 2019 (Raffel et al. 2019).

To train and evaluate these models, the data has been divided into a training, validation, and evaluation set. However, since some protests have substantially more instances than others, a bias in the model arises, often classifying a protest as one of the larger, more prominent topics. To mitigate this, a maximum of 3000 protests per topic is being used. This value has been determined through various experiments with different numbers, with 3000 yielding the most favorable re. The divisions follow the 70-20-10 ratio in attempt to prevent overfitting and to allow the model to change its learning parameters with the validation test, as discussed by Gillis 2024. However, this evaluation approach may introduce bias. The test set consists solely of notes that were previously labeled using the trigger-word method. Consequently, these notes contain explicit trigger words, which the model might learn to associate directly with topics rather than capturing the underlying context. To mitigate this risk, it is crucial to assess the model's performance on notes that have not been labeled and therefore do not contain trigger words. Since these notes lack labels, manual annotation is required for evaluation. We will apply the same sampling methodology by selecting a random sample of unlabeled notes for manual review and evaluation.

We also applied semi-supervised learning to the problem. The implementation can be found in section 5 and the results are in Figure 7 in the appendix.

3.2 Results

3.2.1 Labeled data

Table 1 shows the topics that were labeled using trigger words and how many protests were labeled as that specific topic.

Topic class	Amount of protests
Pandemic	20684
Labor rights	19357
Farmers	10047
Palestine-Israel conflict	7632
Education	7445
Women rights	6514
Ukraine-Russia war	6074
Climate	5551
Environment	4742
Health care	3965
Policies & politics	3633
Lgbtq+	2814
Public services	2620
Immigration	2113
Discrimination	2094
Animal welfare	1749
Black lives	924
Housing	868
Culture	716
Unjust law enforcement	554

Table 1: Amount of protests found per topic using trigger words

It is important to note that during the labeling process, some topics were left out due to a lack of protests. However, it is still insightful to share them. These topics were: prisoner rights, sports, historical justice, anti-crime, Cyprus conflict, Sudan conflict, and military in general. ³

The performance of this classification by trigger-words can be seen in Table 2. Out of 385⁴

³With more specific trigger words more protests might have been found

⁴The calculation of the sample size can be found in the appendix.

randomly chosen rows, 357 have been labeled correctly. This means that, with a confidence rate of 95%, the accuracy is **92.7%** with a margin of error of $\pm 3\%$ (following the formula presented in 3.1.1). Some protests have been marked as "semi-correct". These are protests that have been marked with a topic that does not perfectly fit the note, but is very closely related. Examples of this are when a protest about women's rights has been marked as discrimination or when a protest about salary increase for doctors has been marked as health care. When half of these protests are included as correct in the statistics, the accuracy increases to **94.3%**. When all semi-correct labels are counted as correct, it increases to **95.8%**

Metric	Absolute amount	Percentage
Total	385	100%
Correct Label	357	92.7%
Semi correct Label	12	3.1%
Incorrect label	16	4.2%

Table 2: Performance of classification by trigger words

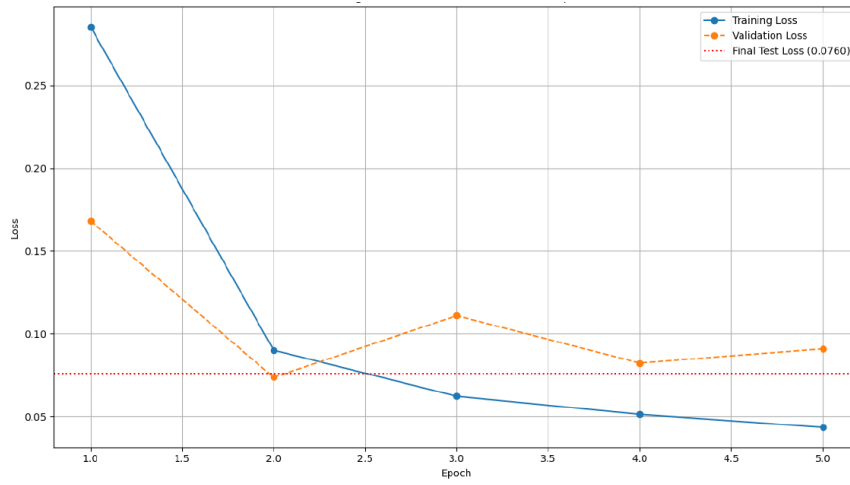
3.2.2 T5

In Table 3, we can find a clear division between the three types of datasets used to train, validate, and test the model.

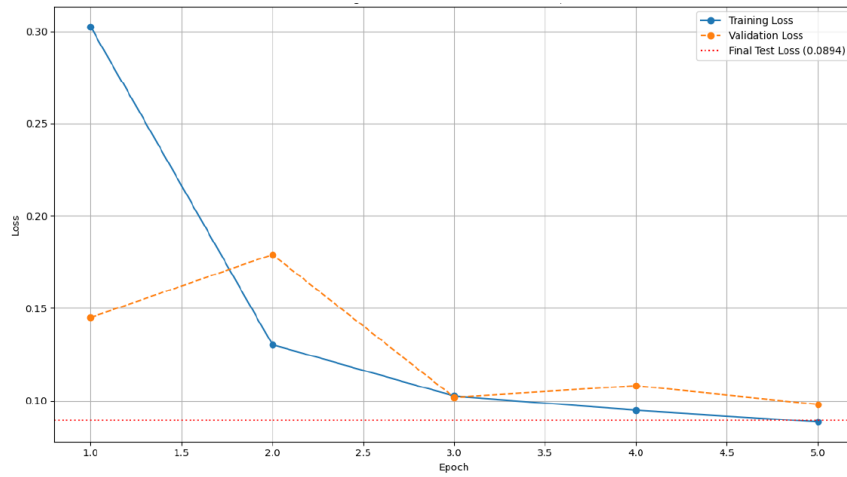
Topic class	Amount of training notes	Amount of Validation notes	Amount of Test notes
Palestina-Israel conflict	2159	576	265
Ukraine-Russia war	2155	583	262
Environment	2129	590	281
Pandemic	2112	572	316
Women rights	2110	582	308
Farmers	2102	600	298
Climate	2101	605	294
Labor rights	2080	601	319
Health care	2078	631	291
Education	2068	632	300
Politics & Policies	2064	613	323
LGBTQ+	1935	611	268
Public Services	1831	500	289
Immigration	1471	420	222
Discrimination	1458	422	214
Animal welfare	1222	362	165
Black Lives Matter	646	189	89
Housing	585	182	101
Culture	513	132	71
Unjust law enforcement	397	87	70

Table 3: Amount of notes per class for training, Validation and test

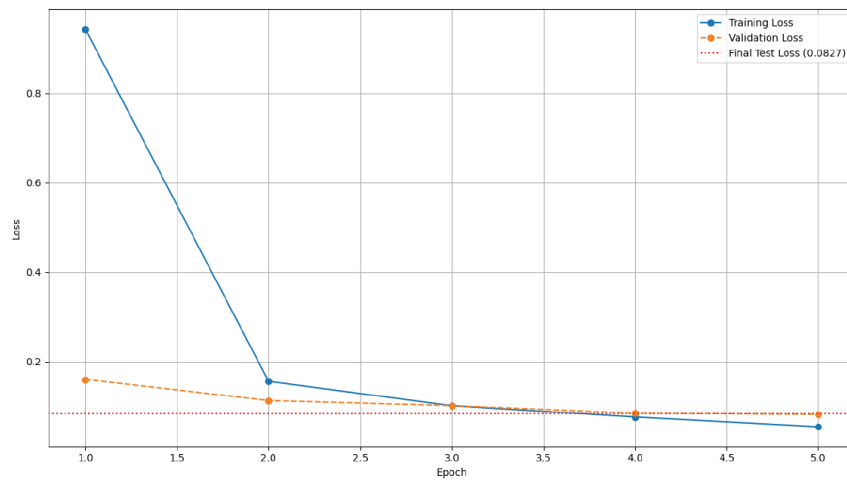
Based on Figure 1, we can see that each model's training loss, along with the validation and test losses, stabilized after 5 epochs. The validation and test values closely resemble each other, which is a good indication that none of the models are strictly only understand the training data. We can also observe that T5 learns the data faster, suggesting it truly doesn't need as many as four epochs for stabilization.



(a) BERT



(b) RoBERTa



(c) T5

Figure 1: Overall training and validation loss for each model

In Table 4 and Table 5, we can observe the performance of the three different transformer models. Based on the metrics, BERT appears to perform the best, closely followed by T5, and then RoBERTa. BERT achieves the highest score on nearly every metric, with the exception of the Average Loss on the validation set, where T5 scores highest. This outcome was unexpected, considering BERT’s earlier development date and its commonly assumed limitations relative to the other two models.

T5 demonstrated a distinct advantage in terms of training efficiency. Whereas BERT and RoBERTa required comparable training durations, T5 completed the process in roughly two-thirds of the time.

Model	Average Loss	Accuracys	Precision (weighted)	Recall (weighted)	F1-Score (weighted)
BERT	0.0909	0.9827	0.9831	0.9827	0.9828
RoBERTa	0.0979	0.9797	0.9802	0.9797	0.9797
T5	0.0812	0.9820	0.9821	0.9820	0.9820

Table 4: Validation set metrics per model

Model	Average Loss	Accuracys	Precision (weighted)	Recall (weighted)	F1-Score (weighted)
BERT	0.0760	0.9848	0.9852	0.9848	0.9849
RoBERTa	0.0894	0.9825	0.9829	0.9825	0.9826
T5	0.0827	0.9827	0.9829	0.9827	0.9827

Table 5: Test set metrics per model

As mentioned before, when evaluating on labeled protests, it is possible that models tend to learn certain trigger words. To really evaluate their generalizability, models should label unlabeled protests (protests with notes without any of the trigger words). These have been evaluated by hand. After evaluation we concluded that T5 performed the best, so the accuracy of this model has been evaluated (other models were not evaluated due to them being significantly worse) in Table 6. Out of the 451⁵ protests, 348 have been labeled correctly. Therefore, with a confidence rate of 95%, the accuracy is **77.2%** with a margin of error $\pm 4\%$. When semi-correct labels are for the half counted as correct, the accuracy would increase to **82.3%**, and to **87.4%** when all semi-correct labels are counted as correct. During the process of evaluating, some protests were skipped since there were no classes for them in our implementation (due to the threshold of 500 for training and in some notes the absence of a topic). However, these protests still got a class assigned that was close in semantics. For example: a protest for a ceasefire in the middle east, which was not about Palestina-Israel, had been classified as palestina-israel war. An example of a semi-correct label is: "On 28 May 2022, vegan activists belonging to Animal Rebellion and Extinction Rebellion gathered at a McDonald’s in Exeter to stage a protest demanding fast-food chains do more to reduce their impact on the climate." being classified as animal welfare instead of climate.

Metric	Absolute amount	Percentage
Total	451	100%
Correct Label	348	77.2%
Semi correct Label	46	10.2%
Incorrect label	57	12.6%

Table 6: Performance of classification (T5)

T5 has been used to classify all protests in the dataset, this final result can be found in Figure 2.

⁵The calculation of the sample size can be found in the appendix.



Figure 2: Final amount of protests per class classified by T5

4 Predicting violence

4.1 Implementation

To find out what factors contribute to violence in protests, multiple predictive models have been made to classify whether a protest will be violent or not.

4.1.1 Preparations

The dataset contains a column specifying sub-event types, which includes four categories: excessive force against protestors, violent demonstration, peaceful protest and protest with intervention. After a careful examination of notes corresponding to each sub-event type, we decided to classify both "excessive force against protestors" and "violent demonstrations" as violent protests. Based on this, we can also conclude that rows marked as either "peaceful protest" or "protest with intervention" are non-violent. We examined the other columns to find potential factors contributing to protests escalating into violence. The two most apparent factors identified were the protest actor (i.e., organizing group or institution) and the country. Country names were removed from actor entries to avoid category fragmentation and ensure generalization by group type only. Without this simplification, the dataset would contain too many narrowly defined actor types. Protests involving multiple actors have been split and treated as individual events per actor for the same reasons. For both countries and actor groups, we recorded the number of violent and non-violent protests. It is also worth noting that additional features, such as time-related variables, were tested but did not improve the model's performance.

4.1.2 ML for binary output

To predict whether a protest will be violent or not, a *logistic regression* model is used. This model is used because it offers a great combination of predictive ability and the ability to see what factors influence violence the most. Using the results of the model, we can use its interpretable coefficients to determine which features result in a protest being classified as violent (Juliana Tolles 2016).

Besides logistic regression, other models are also trained to predict whether a protest is violent or not. These models, however, do not have the capability to analyze what factors influence violence in a protest. The models used are *XGBoost*, *random forest classification*, and *gradient boosting*. These are used because of their ability to capture complex relations between features (Tianqi Chen 2016, Aakash Parmar 2018, Pritika Bahad 2019). These models can be used as a comparison for the quality of the predictions of the logistic regression model.

To train the models, the data is first split into a training set including 80% of the data and a test set using the other 20% of the data, an evaluation is not necessary. This ensures there is enough data to train on, whilst guaranteeing the test data is big enough to evaluate (Gillis 2024). One-hot encoding is being used to transform the non-numerical data into numerical input, for the model to be able to train on. The data is also stratified, meaning that the distribution of violent to non-violent protests is equal in both the training and testing sets (Max Kuhn 2013). This is also done because of the skew in the data, with only around 2.46% of the protests being violent. Another way the model compensates for this skew in the data is by weighing a correctly predicted violent protest 40 times more than a correctly predicted non-violent protest, as can be seen in Figure 3. This makes sure that the skew does not influence the importance of the prediction (Garcia 2009).

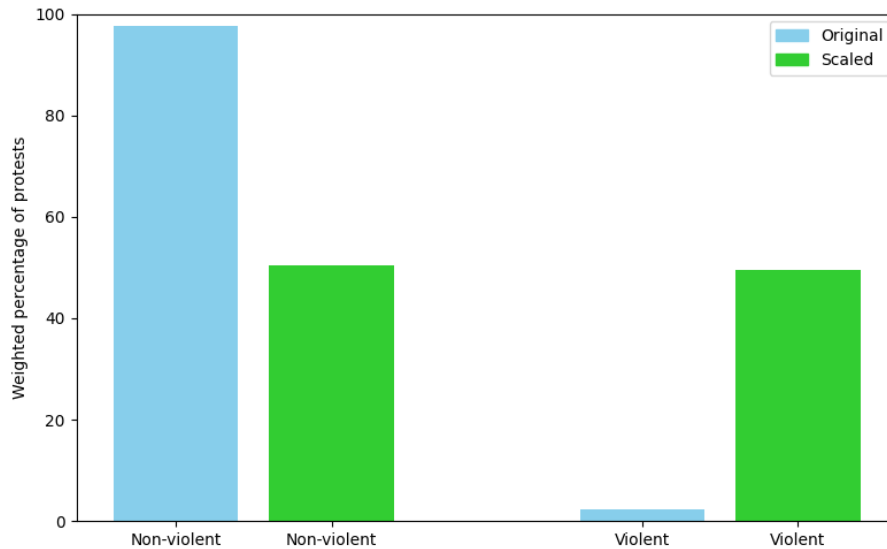


Figure 3: Percentage of violent events by country

The percentage threshold for predicting whether something will be violent or non-violent is based on the maximum value of the F2-score. This is done so more weight is given to recall (B Prasetyo1 2021), which is important to combat the skew in the data. Since there are fewer violent protests, it is important to categorize these correctly, which can come at the cost of marking non-violent protests as violent. Because of this, the models are also evaluated on the F2 score, and by looking at the precision and recall of predicting a protest a violence. Since there is a skew in the data, it is important to look at these statistics to see if the model actually recognizes violence well.

4.1.3 ML for probability as output

Since prior models result in a binary output, another approach we wanted to explore was predicting a probability. This can add nuance and can give additional context regarding the likelihood of a protest turning violent. The probability of a protest escalating into violence given the actor and the country can be calculated using Bayesian statistics. The amount of protests (violent and non-violent) per feature will be used for this.

$$P(\text{violence}|\text{country}, \text{actor}) = \frac{P(\text{country}|\text{violence}) \cdot P(\text{actor}|\text{violence}) \cdot P(\text{violence})}{P(\text{country}, \text{actor})}$$

This formula only works on the assumption that the factors are conditionally independent. However, since an actor can act differently per country, this is an assumption that cannot be made. Therefore, a method including the use of machine learning is required. This has been done by the use of Logistic Regression. Logistic Regression serves a great purpose of assigning a probability to a binary output. Just like the previous models, the same two features, an 80-20 split and a one-hot encoder are being used. Furthermore, explicit metrics are the accuracy and the confusion matrix, but this again shows the binary character. To get an evaluation with more nuance, we used the test set to calculate the average predicted probability of violence for the two different groups: protests labeled as violent and protests labeled as non-violent. By comparing these averages, we can determine how much higher the model's predicted probability is for protests that were actually violent, relative to those that were non-violent. Lastly, for more details, the mean, the median, and the features with their corresponding predicted probability for the highest and lowest probability have been calculated.

4.2 Results

4.2.1 Factors exploration

Initially, two features are employed as predictors of protest violence: country and actor. The proportion of violent events corresponding to each country is visualized in Figure 4. Since there are 2000+ actors, and most actors do not have that many protests, a visualization like this would be deceiving. Because the violent percentages are substantially different per country and actor, these features can be used to predict violence since there is some sort of correlation.

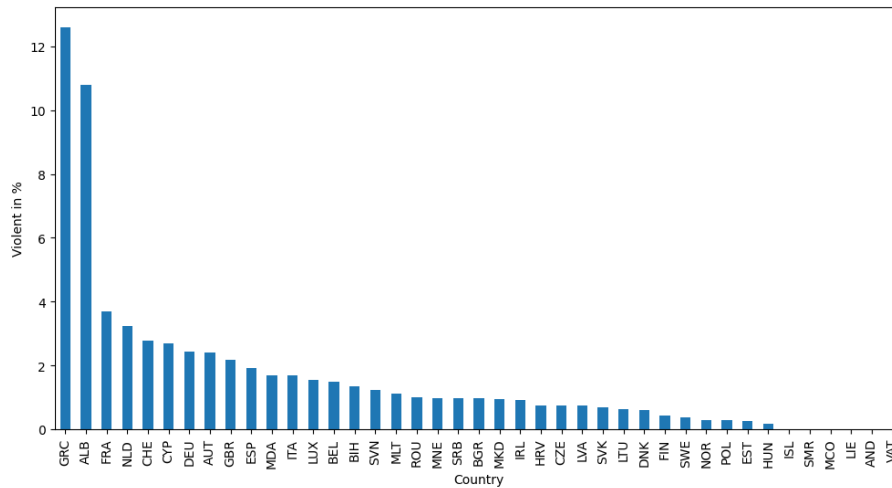


Figure 4: Percentage of violent events by country

4.2.2 Binary prediction

In Table 7, the results of all the models can be seen. Here, the threshold is the probability needed for a protest to be classified as violent. The precision and recall are for predicted violent protests. In Figure 5 the confusion matrices can be seen for the different models.

ML model	Threshold	F2 score	Precision	Recall
XGBoost	0.00	0.228	0.095	0.375
Random Forest	0.51	0.255	0.104	0.389
Gradient Boosting	0.04	0.268	0.114	0.399
Logistic Regression	0.63	0.266	0.103	0.435

Table 7: Violence prediction results per model

	Predicted: Non-Violent	Predicted: Violent
Actual: Non-Violent	69288	3657
Actual: Violent	1189	523

(a) XGboost

	Predicted: Non-Violent	Predicted: Violent
Actual: Non-Violent	67657	5288
Actual: Violent	1059	653

(b) Random forest

	Predicted: Non-Violent	Predicted: Violent
Actual: Non-Violent	67649	5296
Actual: Violent	1033	679

(c) Gradient boost

	Predicted: Non-Violent	Predicted: Violent
Actual: Non-Violent	66521	6424
Actual: Violent	967	745

(d) Logistic regression

Figure 5: Confusion matrices per model

It is possible to look at the features that the logistic regression model finds important when classifying a protest as violent using the coefficients. In Table 8, the 10 features with the highest coefficients are displayed.

Feature	Coefficient
Actor - Rioters	4.75
Actor - Prisoners	4.40
Actor - Black bloc	3.94
Actor - PA: Palestine Action	3.82
Actor - Civilians	3.43
Actor - Shut the System	3.42
Actor - No TAV Movement	3.39
Actor - MiB: Men in Black	3.30
Actor - AUR: Alliance for the Union of Romanians	3.16
Actor - Anarchist Group	3.14

Table 8: Top 10 Features contributing to predicted protest violence

4.2.3 Probability prediction

The model achieves an overall accuracy (on the 20% test set) of 97.95%, as shown in Table 9. Although this accuracy seems good, it does not fully capture the performance nuances. This is especially the case due to the skewed dataset, and the probabilities still being rounded to 0 and 1.

	Predicted: Non-Violent	Predicted: Violent
Actual: Non-Violent	63774	38
Actual: Violent	1298	45

Table 9: Confusion matrix

It is important to understand that a misclassification does not necessarily indicate poor performance by the model. The model outputs a probability of violence, which is subsequently converted into a binary classification using a threshold of 0.5, a simplification that this study aimed to avoid. A protest might be given a violence probability of 40%, but this would still be counted as non-violent, while there is a significant chance of the protest being violent. This is especially relevant given that the model is based on only two features. These features alone are not explicitly predictive of violence in a binary matter, because a pair of a country and actor are not necessarily always violent or always non-violent. An actor could have organized both a violent and a non-violent protest in the same country. Logically a probability of 50% would be desirable, not a binary prediction. For more nuances, other probabilities have been found to give more context of the output of the model. They have been summarized in Table 10

Metric	Probability of violence
Mean for features that were actually violent	7.72%
Mean for features that were actually non-violent	1.91%
Mean violence probability	2.04%
Median violence probability	1.33%
Pair (Greece & Rioters) resulting in the highest probability	98.99%
Pair (Norway & Ukranian Group) resulting the lowest probability	0.24%
Practical example (The Netherlands & XR: Extinction Rebellion)	0.94%

Table 10: Additional statistics for context

5 Documentation

Link to our online repository, with readme.md file included: [Github Repository](#)

6 Conclusion

6.1 Conclusions protest labeling

It can be derived from the results of our various models, that a supervised model is the best fit for this task. This conclusion is based on the accuracy scores. In this report, we tested three different transformer models to increase our chances of success. All three models achieved high performance scores on the labeled dataset. Based on the data we collected on the data, we can conclude that the T5 model is the best fit among the three, since the model shows high metric scores and demonstrates that it is not overfitting on the higher-count classes. The other two models, however, have significant difficulty with the dataset and overfitting on the trigger words, which causes them to predict classes like "culture" more frequently than "women's rights". The results on unlabeled data is not satisfying yet, but it does not show that the product itself does not work. Its potential could be achieved by using a different implementation, which includes either lowering the threshold or finding more protests with another topic (which can lead to more classes).

Also by testing by hand, we indicated that all three models had problems with subjects that were introduced beyond 2019. Subjects like Covid-19 (a deadly virus in 2020) were not part of the pre-training data that the researchers of the models used, so classes like "pandemic" were hard for the models to classify.

6.2 Conclusions violence predictions

As can be seen in the results section, the binary classification models seem to struggle with accurately predicting whether a protest will be violent or not. This is reflected by the low F2 scores of all the models, with the highest score being 0.266, indicating it poorly identifies the minority class, being violent protests. The precision and recall for predicting a protest as violent are also low. The logistic regression model has a precision of 0.103, indicating that the model too often predicts that a protest will be violent while, in reality, it will not be violent. This is an aftereffect of choosing to maximize the F2 score, which focuses on the recall. This can be seen in the results of the recall being 0.435, meaning the model finds the positive cases fairly well. However, with the precision being so low and the recall being better but not reliable enough, it can be concluded that this model is not reliable.

The average predicted violence probability for actual violent protests is around four times higher than for non-violent protests. This shows that the model indeed captures certain signals leading to violence, despite the imbalanced dataset. It can be concluded that the probability model is less limited than the binary classification. It gives more nuance in the form of gradations of percentage, which is especially useful for prioritization and as an estimation for a level of violence.

7 Discussion

7.1 Limitations

There are some limitations to our product. Starting with the topic labeling; during this project, we have encountered many protests that are not explicitly about one specific topic. When farmers or teachers protest for their sector, this is very closely related to policies as well. And when teachers and health workers protest to get better loans, this is both about labor rights as it is about education/health care. So the limitation itself is that our product does not offer multi-class labeling.

The models that labeled protests with a topic, are limited to labeling with protest topics they were given. Due to our threshold and taking into account the duration of this project, some protest topics that are a noteworthy size of the total amount of protests are not in our models.

Due to potential overfitting, we included a method by taking a sample size to evaluate protests that were not labeled by trigger words. The (hyper)parameters, however (like the amount of epochs that they were trained on), were finetuned based on the labeled protests. This means that the hyperparameters have not been adjusted based on generalizability.

Due to the outdated dataset used by the pre-training models, which made them less effective in classifying classes like 'pandemic' and consequently decreased the accuracy of their predictions, there is a need for a transform model that is pre-trained on newer data from beyond 2024.

7.2 Ethics

It is important to note potential ethical issues from the product. The feature "country" that is being used to predict violence may raise ethical concerns. This feature can enhance harmful stereotypes and other biases. Especially since historical data is used without other broader topics like socio-economic and political contexts that contribute to certain outcomes. Also, the supervised models have been pre-trained before. There could be a risk of any biases that we are not aware of.

7.3 Future research

The labeling that has been done for the training set is based on trigger words. This approach was due to the time constraints of the project. But this means that the model was trained on protests that were not marked perfectly. So in future research, it is recommended to have them labeled by hand or by a reliable language model. This could potentially lead to fine-tuning the language model more accurately.

In light of the earlier discussed issue of the supervised labeling model not being able to be updated to include more topics. This could be improved upon by using a semi-supervised or unsupervised model, since both are easy to update. The attempted models in these categories are less than optimal. However, we believe that with better hyperparameters or with a larger model, these results could be improved.

The (hyper)parameters have been finetuned based on the labeled protests. We have seen in the results that two of the supervised models tend to overfit and are not able to generalize well. Due to time constraints, we were not able to fix this, because evaluating the unlabeled topics by hand is very time-consuming. In future research, this could be prevented such that the other models do not overfit and a final model could be chosen more "fairly." Also, T5, which does not tend to overfit, could be improved this way.

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A Appendix

```
{
"blm": ["black lives matter"],
"lgbtq": ["lgb", "lesbian", "gay", "homosexual", "transsexual", "queer", "homophobia", "trans-
phobia", "biphobia", "trans rights"],
"women rights": ["women's rights", "feminism", "feminist", "against women", "women
protested", "abortion", "sexual violence", "sexual assault", "sexual harassment", "sexual
abuse"],
"immigration": ["migrants", "immigration", "against migration", "deportation detention"],
"unjust law enforcement": ["police brutality", "criminalize protests", "criminalize demonstra-
tions", "police misconduct", "police repression"],
"discrimination": ["discrimination", "racism"],
"climate": ["climate change", "fossil fuels", "greenwashing", "climate agenda", "global warm-
ing"],
"palestine-israel conflict": ["gaza", "palestine", "israel", "hamas", "palestinian"],
"animal welfare": ["species extinction", "animal welfare", "animal rights", "labour conditions",
"animal protection", "bullfighting", "animals locked", "wildlife", "cruel"],
"farmers": ["farmers", "agriculture", "agricultural", "intensive farming"],
"labor rights": ["labor agreement", "wages", "rights of workers", "labor rights", "higher
salaries", "working conditions", "commission fees", "pension", "salary equalization", "unfairly
dismissed"],
"health care": ["healthcare", "hospital", "hospitals", "nurseries", "emergency clinics", "emer-
gency care"],
"environment": ["environment", "pfas", "nitrogen", "planned felling", "biodiversity", "park
project"],
"public services": ["collapse of a concrete canopy", "canopy collapse", "bus", "traffic acci-
dents", "railway station", "bike lanes", "road connection", "public service", "pedestrianization",
"child-safe intersections", "bike street", "play street", "reasonable mobility", "cycling condi-
tions", "urban development", "free transport"],
"ukraine-russia war": ["russia", "ukrain"],
"housing": ["residential complex", "dignified housing", "evict"],
"culture": ["tourism", "tourists", "cultural sector"],
"policies & politics": ["social welfare", "social services", "social assistance", "economic justice",
"economic sovereignty", "economic independence", "adoption of the euro", "euro adoption",
"council's plan", "nightlife noise", "municipality", "regional government", "political criticism",
"political opposition", "against the pm", "resignation of the president", "political rights", "po-
litical prisoners", "anti-eu", "pro-eu", "democratic", "referendums", "urgent elections", "distinct
autonomy"],
"pandemic": ["pandemic", "covid", "coronavirus"],
"education": ["education", "teacher", "academic", "education", "professor", "university", "stu-
dent loan"]
}
```

Figure 6: Final dictionary of topics and trigger words used for classification for model training



Sample size calculation for the evaluation of the assigned topics by trigger words:
 $Z = 1.96$ (Because the confidence rate is 95%)
 $p = 0.9$ ⁶
 $e = 0.03$
Filling in these values into the formula gives:

$$n = \frac{1.96^2 \cdot 0.9 \cdot 0.1}{0.03^2}$$
$$n = \frac{3.8416 \cdot 0.09}{0.0009} \approx 385$$

⁶Based on experience by the process of finding notes with a wrong topic.

Category	Precision	Recall	F1-score	Support
animal welfare	0.82	0.81	0.82	1336
climate	0.56	0.02	0.04	5569
culture	0.29	0.43	0.35	1508
discrimination	0.11	0.63	0.18	763
education	0.61	0.68	0.64	8468
environment	0.33	0.81	0.47	4483
farmers	0.95	0.91	0.93	9668
health care	0.25	0.63	0.36	5174
housing	0.17	0.92	0.29	820
immigration	0.21	0.32	0.25	720
labor rights	0.91	0.30	0.46	36309
lgbtq	0.70	0.52	0.60	1681
palestine-israel conflict	0.96	0.71	0.82	7425
pandemic	0.75	0.57	0.65	14804
policies	0.13	0.36	0.19	5598
public services	0.11	0.34	0.17	2988
ukraine-russia war	0.96	0.58	0.72	6169
unjust law enforcement	0.33	0.69	0.44	2176
women rights	0.94	0.40	0.56	6835

(a) ZSC

Category	Precision	Recall	F1-score	Support
animal welfare	0.70	0.89	0.78	1336
climate	0.72	0.81	0.76	5569
culture	0.29	0.21	0.24	1508
discrimination	0.06	0.57	0.10	763
education	0.73	0.33	0.45	8468
environment	0.59	0.53	0.56	4483
farmers	0.91	0.91	0.91	9668
health care	0.25	0.19	0.21	5174
housing	0.12	0.88	0.21	820
immigration	0.22	0.83	0.34	720
labor rights	0.95	0.39	0.55	36309
lgbtq	0.64	0.64	0.64	1681
palestine-israel conflict	0.95	0.82	0.88	7425
pandemic	0.38	0.92	0.53	14804
policies	0.16	0.04	0.06	5598
public services	0.20	0.02	0.04	2988
ukraine-russia war	0.80	0.87	0.83	6169
unjust law enforcement	0.40	0.22	0.29	2176
women rights	0.83	0.83	0.83	6835

(b) ESS

Figure 7: Classification reports per model for each category

Sample size calculation for the evaluation of the assigned topics by T5 on unlabeled data:

$Z = 1.96$ (Because the confidence rate is 95%)

$p = 0.75$ ⁷

$e = 0.04$

Filling in these values into the formula gives:

$$n = \frac{1.96^2 \cdot 0.75 \cdot 0.25}{0.04^2}$$

⁷Based on experience by the process of finding notes with a wrong topic.



$$n = \frac{3.8416 \cdot 0.1875}{0.0016} \approx 451$$
